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1. Purpose

To accomplish our world-class quality vision, Ceer is committed to forming a strong partnership with its supplier base. To achieve the expectations of Ceer and “Our Customers” the suppliers shall establish and maintain a comprehensive Quality Operating System, to assure compliance with the requirements of the contract and this document.

The Purpose of this Supplier Quality Handbook is to outline the minimum quality requirements for the Suppliers of Ceer in accordance with IATF 16949. The suppliers must carry out Advanced Product Quality Planning (APQP) and Production Part Approval Process (PPAP) for all series production, as outlined in IATF 16949. Suppliers not having achieved IATF16949 must submit a waiver form and plan to acquire the certification that will be used in the sourcing process.

Unless otherwise specified in the following, the requirements outlined in this process are orientated towards the most recent standards as outlined by the Automotive Industry Action Group (AIAG), specific to the Production Part Approval Process (PPAP) Manual and Advanced Product Quality Planning (APQP) Manual 3rd Edition & Control Plan 1st Edition released by AIAG.

This Supplier Quality Handbook, however, is not intended to change specific requirements, which in case of inconsistencies, the AIAG standards shall take precedence.

2. Scope

This handbook applies to all suppliers, who must ensure that the Ceer requirements, for all vehicle subsystems, are understood and implemented along the whole Supply Chain.

These requirements are designed to assure that all products (systems, modules, components, bulk materials, etc.) supplied to Ceer meet all known drawings, standards, specifications and agreements. Meeting these minimum requirements will ensure customer satisfaction.

The term “shall”, as used in this document, indicates mandatory requirements. The word “should” indicate a suggestion or recommendation.

3. Acronyms and Abbreviations – Ceer Specific

Refer to Appendix A



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4. Responsibilities

It is the responsibility of the supplier to ensure all the required products are delivered on time, to the specified quantity and in conformance with the quality, performance, and engineering specifications.

The supplier shall communicate with Ceer if any supply chain delivery, quality, or performance issues arise, and take adequate countermeasures to contain and mitigate such issues.

Suppliers shall leverage their expertise and support Ceer by providing feedback, highlighting concerns, Things Gone Wrong (TGW), sharing lessons learned, and suggested improvement actions in order to enhance overall quality, reduce cost, scrap, and improve throughput, among other.

As early as possible in the program, the Supplier Senior Management shall:

- Communicate to Ceer their current organizational structure and individual contacts.
- Define the individuals responsible for carrying out Advanced Product Quality Planning (APQP).

The designated individuals performing those functions shall have sufficient expertise, well-defined responsibility, authority, and organizational freedom to identify, evaluate and resolve quality problems that may arise.

The Supplier Senior Management shall communicate the above requirements to Ceer by completing and returning the Supplier Contact & Open Issues Tracking Program Specific (CEER-SQ-QM-001-F003) form.

Any changes to the organizational structure and APQP core team shall be communicated to the Ceer Purchasing Buyer and Supplier Quality Specialist.

The official language for communication is English unless specified and agreed mutually by Ceer and supplier. All documents shall be provided in English and shall refer to the metric system, except otherwise stipulated in the Handbook.

The supplier is responsible for ensuring the availability of personnel who can communicate effectively in English to facilitate efficient communication regarding all technical, quality, or any other APQP-related topics.

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5. Product Development Process

Ceer Product Development follows a series of processes/procedures defined as the Ceer Supplier Development Process. This details the methods and tools used by Supplier Quality from the initial assessment as a potential new supplier / location through launch and into serial production with tactical monitoring within operations.

APQP framework exhibits standardized quality requirements, which enables suppliers to design, develop and produce a product that meets Ceer requirements.

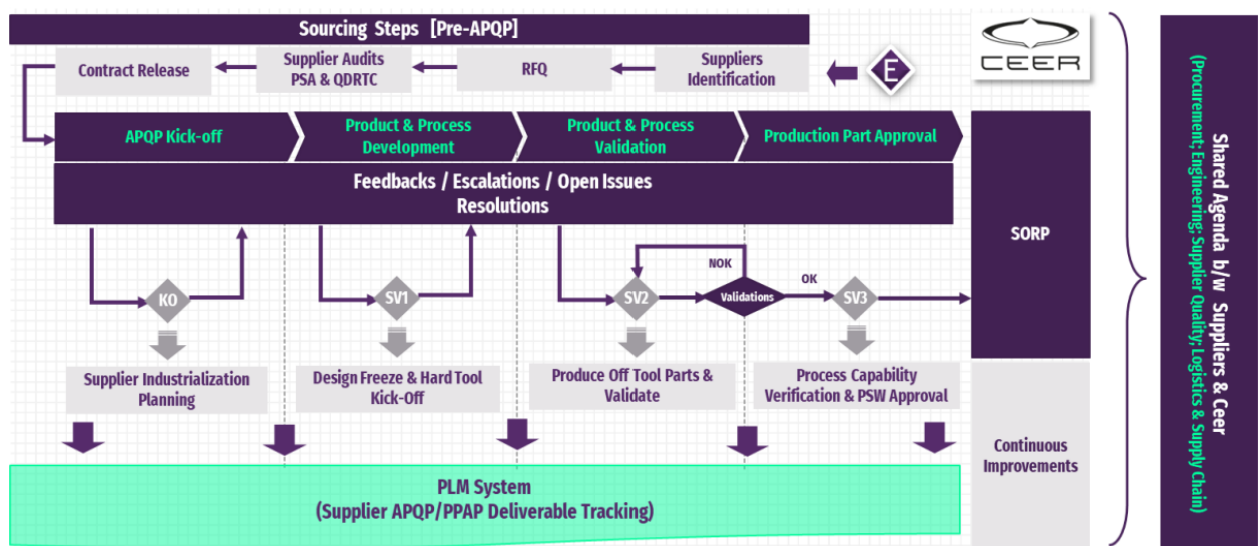


Fig.01. Ceer APQP Process Definition

Ceer APQP is a synchronization of product and process development activities and ensures communication internally and externally.

The process of Advanced Product Quality Planning (APQP) at Ceer involves four key events, each contributing to the successful development and launch of a product. These events are organized in a sequential order: KO, SV1, SV2, and SV3.

KO (APQP Kick-off - Supplier Industrialization Planning):

The journey begins with the APQP Kick-off, a crucial milestone in the product development process. During this stage, teams gather to initiate the project and set the groundwork for its success.

Supplier Industrialization Planning takes place, focusing on how suppliers will integrate into the production process seamlessly.

SV1 (Product and Process Development - Design Freeze, and Hard Tool Kick-off):

SV1 represents a critical phase where product and process development are undertaken. Teams work together to refine the product design and align it with production capabilities. The stage culminates with the important milestones of Design Freeze and the initiation of Hard Tool manufacturing, marking a significant step forward.



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SV2 (Product and Process Validation - Produce Off Tool Parts, and Complete Validations):

SV2 focuses on validating the product and production processes. Off-tool parts are manufactured to ensure they meet quality standards. Comprehensive validations are carried out to confirm that the product and process are ready for serial production.

SV3 (PPAP Approval - Process Capability Verified, and PSW Approved):

The final stage, SV3, involves the submission of the Production Part Approval Process (PPAP) for approval. Process Capability is thoroughly verified to ensure it meets required standards. The Production Part Submission Warrant (PSW) is approved, indicating that the product is ready for full-scale production.

These four APQP events are strategically aligned to guide the product development process from inception to production, with a strong emphasis on quality assurance and collaboration among teams and suppliers.

6. International Material Data System (IMDS)

To ensure compliance with the various legal and market requirements, Ceer requires its Suppliers to report material and substance information for all types of purchased materials, components or items supplied to Ceer. All substances and/or materials shall be reported to Ceer using the International Material Data System (IMDS) (www.mdssystem.com).

Suppliers shall submit the required IMDS to Ceer as soon as possible upon award of new business, but always as part of the PPAP process. The supplier IMDS information shall be subject to Ceer review and approval. Once approved by Ceer, the Supplier of the material or component shall indicate such approval in the PPAP documentation supplied to Ceer regardless of submission level requested.

The supplier shall also implement procedures or controls necessary to prevent the introduction of prohibited and restricted substances in materials as specified herein into the final product and/or component supplied to Ceer.

7. Ceer Requirement

Ceer supply management requirements are based on four key processes: supplier selection, new product launch, continuous improvement, and supplier rapid issue containment and resolution. These key processes are global in nature as are any of the referenced procedures.

I. Criteria for selection as a Ceer Supplier

i. IATF 16949 Registration

The goal for all Ceer suppliers of materials and services affecting production material is to demonstrate compliance to IATF 16949. Suppliers shall also comply with Ceer specific requirements defined in the Ceer Supplier Quality Handbook (CEER-QUSQ-SP5-L2-001).

Suppliers to Ceer shall have a plan to achieve conformity to IATF 16949. Unless otherwise specified, conformity is demonstrated by third party certification to ISO 9001 (at minimum) or IATF 16949. This is consistent with the expectations of Ceer business system that complies with IATF 16949 requirements. The scope of the requirement affects subassembly, sequencing, sorting, re-work and calibration services in addition to direct material suppliers.



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Ceers has a special exemption policy for IATF 16949. During the supplier registration process, an IATF Waiver Deviation Request Form (CEER-SQ-QM-001-F026) shall be submitted by supplier and approved by Ceers Supplier Quality.

Additionally, suppliers shall maintain and provide certification status once per year. Suppliers shall immediately communicate any change in certification or status to their respective Supplier Quality Specialist. Suppliers and sub-suppliers identified as special process providers are to adhere to the to the AIAG Special Process requirements.

Ceers requires that its suppliers use the latest Automotive Industry Action Group (AIAG) version of the Advanced Product Quality Planning (APQP), Control Plan (CP), Potential Failure Mode and Effects Analysis (FMEA), Measurement System Analysis (MSA), Production Part Approval Process (PPAP), Statistical Process Control (SPC) and Machinery FMEA manuals as guidelines for their system development. For these publications, visit <http://www.aiag.org>.

ii. Environmental and Sustainability Compliance

Ceers suppliers are expected to fulfil the sustainability requirements outlined in the General Terms and Conditions (GTC)

Our suppliers are required to develop internal company policies, to obtain ISO14001 and ISO45001 certifications, and engage in periodic evaluation, planning and training required to ensure compliance to the sustainability requirements.

iii. e-Business Capabilities

Suppliers shall have email, internet access and internet browser as a minimum for e-Business capability. This is required to participate in Ceers web bases applications and communications. These include but are not limited to:

- Supplier Portal (self-registration limited to three supplier personnel)
- Supplier Request for Change, eForm
- Supplier Performance Scorecard
- Supplier Chain Portal / Ordering
- Supplier Corrective Action System, eForm
- Supplier Audit Data Base
- Supplier APQP / PPAP Submission. eForm

Suppliers are responsible for maintaining contact information in those e-Business systems. The minimum information required includes contact name, title, office, mobile phone and email.

These individuals are expected to be available 24 hours, 7 days a week to handle logistics issues and respond to quality issues.

These activities may include a rapid response to items such as maintaining supply of components and/or subsystems, initiating supply and/or quality issue containment, providing issue root cause, and permanent corrective actions in a timely manner.



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II. Supplier Quality Roadmap

i. Supplier Sourcing and Early SQ Engagement

This phase initiates before awarding a Supplier. It requires the participants of Buyer, SQS, and Engineering to evaluate potential Suppliers fully from Technical, Commercial, Quality and Process.

Process initiates with design concept, initial specification, and forecast demand provided to Purchasing. Information is reviewed by Buyer and SQ (for early engagement). List of potential Suppliers are defined.

Potential suppliers are contacted by Supplier Quality Specialist to coordinate a PSA

(Potential Supplier Assessment) as per CEER-SQ-QM-001-F002 master form. The PSA result is reviewed internally with a Supplier Development plan created as warranted. Ceer suppliers under a Supplier Development plan shall complete all actions ahead of the Ceer Manufacturing Validation Build – Non-Saleable (MVB-nS) (or) end of SV2 event.

8. Advanced Product Quality Planning (APQP)

I. Introduction

Ceer uses Advanced Product Quality Planning (APQP) as Quality Project Management Process.

It enables Ceer to monitor a Supplier's progress of component supply from Sourcing through to Production.

Ceer APQP process (CEER-QUSQ-SP5-L2-002) contains a set of actions and activities, which when completed will ensure component delivery to Ceer for specific build phases, with the correct component(s) maturity and installed manufacturing capacity.

APQP will be undertaken during all Projects by Ceer with all Suppliers that have been successfully sourced for a new Project. It may also be used to manage change content in Model Year Updates for current production Series.

APQP commences with the supplier within 30 days of Ceer Purchasing nomination and continues +90 days of SORP.

APQP will continue until the Supplier has provided 100% of component(s) to Full PPAP Maturity to Ceer.

During the duration of the Project, a Core Team of people must be assembled comprising of various members of different departments from Ceer and the Supplier.

The Core Team will normally comprise of:

- Ceer SQ Specialist
- Ceer Buyer
- Ceer Project Engineer
- Ceer Supply Chain Engineer (Logistics/Packaging)
- Supplier's nominated representatives (Project Manager; Quality; Engineering; Logistics)

The APQP Process contains several steps which will be defined below.



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This handbook is intended as a guideline on proceeding with all steps of APQP. Ceer periodically reviews and revises this handbook. It is recommended that this handbook is placed under the Document Control procedure of the supplier and issued to the relevant departments within the supplier's organization.

9. Quality Activities Prior to Production Part Approval

I. Technical Reviews

Lead by the Ceer Engineer, this activity reviews the supplier's technical proposal, to ensure that all requirements in the RFQ package have been understood and the supplier has a plan to produce parts meeting the Ceer expectations. The aim of this review is to identify capable suppliers which meet the Ceer expectations.

Both the Ceer Product Development Engineer and Supplier Quality Specialist should identify agenda items to be reviewed in the technical meeting and assess the supplier's technical, manufacturing and quality capabilities for continuation in the sourcing process.

The supplier's understanding of all part and process specific requirements is assessed, to ensure these have been considered in its proposal.

II. Sourcing Eligibility

The SQS carries out the Potential Supplier Assessment (PSA) activity, to determine if a supplier's manufacturing location is eligible to receive business from Ceer.

Upon the PSA activity completion, the SQS ensures that Supplier Development plans are created for all potential suppliers where gaps are identified. It is the supplier's responsibility to create these plans and complete all improvement actions at the earliest time possible.

The SQS shall verify the progress periodically and drive the supplier to complete all activities before PPAP submission / approval.

III. APQP Kick-off Meeting and Supplier Verification (SV)

The APQP Kick-Off meeting shall be led by the Ceer SQS and subsequent APQP meetings shall be held on a regular basis, on a minimum cadence of one per month, and led by the supplier.

The APQP Kick-off meeting activities shall ensure as minimum, the following:

- Confirm APQP Core team members, substitutes, and escalation contacts.
- Determine meeting cadence and locations.
- Review Supplier Development Plan.
- Review current Open Issues List and document any additional points.
- Confirm volume and daily contracted capacity.
- Review Program Timing and milestones.
- Review quality deliverables for all major Program Milestones.
- Review gauges, tooling, and equipment requirements.
- Agree on supplier verification (SV) audit timing based on Program Milestones



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The SQS is responsible for adequate documentation of the APQP Kick-off meeting, and the supplier is responsible for the adequate documentation of all subsequent APQP meetings.

Although the agreed meeting cadence is decided and agreed upon by the entire APQP team, it is recommended that the APQP meetings are held on a weekly basis.

Clear meeting minutes and updated APQP Open Issue List are expected to be issued out to the APQP Core team no later than the Close of Business of the respective day.

IV. Timing Charts / Open Issues

The supplier shall create and keep updated APQP timing charts and Open Issues List. These documents will be reviewed in detail during the APQP meetings, to ensure that program deliverables are met on time.

The supplier shall communicate immediately to the SQS any changes in timing, concerns or issues that might impact on the Ceer program timing. It is expected of the supplier to create and action recovery plans, to mitigate these newly identified risks.

The SQS shall review supplier's APQP timing plans and ensure key activities and tasks are considered, including any sub-tier component development timing.

It is recommended that all sub-tier component milestones are planned to be completed at least 6 weeks earlier, to ensure Ceer program timing is protected.

V. DFMEA

This is a living document lead by the design owner; either the Ceer engineer or the Supplier (if supplier is design responsible).

Ceer can provide the supplier with customer input of functions and requirements, cascaded from the CEER system FMEA's, which must be included as minimal content in the design requirements. Preference is for supplier to adopt the Ceer System & Design FMEA Procedure to downstream activities; however, the use of SAE J1739 or AIAG VDA standards is supported.

This document is continuously updated to include any new information of changes occurred during product development.

The SQS will confirm if the supplier (if not design owner) and Ceer engineer are working together on the development of the DFMEA, and the necessary information is available as input into the PFMEA.

If not, appropriate action will be taken to ensure this team activity is initiated.

Recommended action must be made for any High Action Priority items to reduce them to minimum of Medium ratings and documented into DFMEA. Refer to the AP rating charts in the Ceer System & Design FMEA CEER-TEPI-COP2-L2-001.

The supplier shall provide any lessons learnt to support the development of the DFMEA.



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VI. Design Reviews

Reviews to ensure the design has been defined to enable construction of tools and gauges.

Design owner schedules and conducts workshops to define the critical product and process control characteristics.

The design owner will also schedule and conduct GD&T reviews to define the dimensioning and tolerances on the drawing (3D models).

The SQS and supplier are supporting these reviews to:

- Buy-off the GD&T structure.
- Agree which special characteristics can be measured with attribute gauges.
- Confirm if the current manufacturing process can achieve the specified tolerances.

Any concerns relative to the special characteristics or GD&T or manufacturability of the part need to be documented on the APQP open issues list by the supplier.

The supplier shall ensure all special characteristics, GD&T or product and process specific requirements are included in the tooling, gauges and equipment.

VII. Gauges, Tooling, and Equipment Review and Approval

The gauge, tooling and equipment reviews start at the APQP Kick-off meeting. The purpose of these reviews is to ensure the manufacturing process is designed to the latest part engineering level and it is capable of producing parts to the quality and rate required by Ceer Program requirements as follows:

The APQP Team, led by the SQS, will review:

- Tooling; gauges and testing equipment concepts.
- Component functional testing concept and criteria.
- Matching Build strategy, requirements, and data collection.

This is to ensure that the supplier manufacturing process can meet component design intent and required process capability. Tooling and equipment reviews will be held throughout the building process as part of the APQP meetings and Supplier Verification (SV).

The supplier is required to ensure all tooling, gauge and equipment is designed to the latest Ceer engineering design level.

A detailed timing chart shall be submitted by the supplier for all tooling, gauges and equipment. At minimum, the supplier shall consider the following activities within their timing plan:

- Ceer Engineering Release.
- Manufacturing feasibility assessment.
- Tools and gauges concept and design finalization.
- RM ordering & arrival at tool maker for milling.
- Tools & Gauges KO at tool maker.
- Assy line KO at production supplier location (If any assy. process required).
- Tools and gauges construction and try-out (at tool maker)



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- Tools and gauges buy-off at final production location.
- Assy line tryouts/ Machine capability verification.
- Gauge certification and MSA validation.
- PV / AAR & Matching completion & signoff.
- PPAP approval.

The supplier shall inform the Ceer SQS of any changes to the timing submitted to Ceer APQP team and develop reaction plans to mitigate any risks to Ceer program timing.

Upon Tooling Kick-off (TKO) approval from Ceer SQS, the supplier can proceed to tooling, equipment and gauge manufacture and certification.

The Ceer SQS will review, for approval, gauge accuracy, instructions, and integrity for fit and function and MSA. It is recommended for the supplier to complete the AIAG APQP New Equipment Checklist (A-3) for review with the Ceer SQS.

VIII. Process Flow Charts

Upon sourcing completion and APQP Kick-off, the supplier shall define a production Process Flow Chart.

The production process flow chart shall include all items part of the manufacturing process, respecting the nomenclature (move, inspect, store, etc.)

The supplier is expected to complete this activity as per APQP timing plan and submit the process flow chart to SQS for review, including a completed checklist A-6, found in the AIAG Advanced Product Quality Planning and Control Plan manual, Appendix A, Process Flow Chart Checklist A-6.

The process flow chart will also be reviewed by SQS at the SV assessments to ensure completeness and compares with the production line. The SQS will check to ensure that the flow chart is linked to the PFMEA and Control plan and walk the manufacturing line to ensure all steps are completed. The supplier shall initiate PFMEA team meeting, prior to sourcing, and involve SQS, Ceer Engineer, Manufacturing Engineer, to provide a preliminary PFMEA.

IX. PFMEA

The Supplier shall prepare the PFMEA document, cross-functionally, involving all required internal teams. For supplier PFMEA development, Ceer supports the use of SAE J1739 or AIAG VDA standards.

Any changes or recommended improvements actions shall be communicated to the Ceer APQP team.

At Minimum, the supplier shall input in the PFMEA development:

- DFMEA
- Warranty Data
- Build Concerns
- Inhouse rejection analysis (scrap, FTQ, etc.)
- Ceer Lesson Learned ; Supplier Lesson Learned
- Process Flow Diagram / Structure Tree



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Recommended action must be made for any High Action Priority items to reduce them to minimum of Medium ratings and documented into PFMEA. Refer to the AP rating charts in the Ceer System & Design FMEA CEER-TEPI-COP2-L2-001.

It is recommended for the supplier to complete the AIAG APQP Process FMEA Checklist (A-7) for review with the Ceer SQS.

X. Control Plan

The supplier is expected to develop a Pre-launch Control Plan, to outline the additional process controls for all features, additional inspection and testing during the early build phases, while the product is not supplied from the final production process and location.

This is used to validate the Production Control Plan and not as a substitute for it. These documents shall be prepared in accordance with the AIAG Control Plan (CP) 1st Edition manual, and it shall include all process steps, inspection points and all critical and significant product / process characteristics.

Any modifications to the Control Plan documents shall be reviewed for approval with the Ceer SQS.

It is recommended for the supplier complete the AIAG Control Plan Checklist (A-10) for review with the Ceer SQS.

XI. Process Audit

The process audits consist of a set of verification activities, that assure robust quality processes are established at supplier's manufacturing location.

These verification activities cover various suppliers' internal processes such as: Non-conforming Materials; Layered Audits; Error Proofing; Red Rabbit / Dummy Test Final Inspection; Problem Solving; FIFO; Process control and Process capability, Training Plan etc. It is at the discretion of Ceer SQS to request additional verification activities or to reduce them, on a case-by-case basis.

This activity is led by the supplier and monitored at the discretion of Ceer SQS.

This activity shall be completed prior MVB-nS build phase and all following serial home line production build events. All issues and actions identified shall be included on the APQP Open Issues List.

The Ceer SQS shall monitor the supplier action plan and ensure all actions are closed in a timely manner.

XII. APQP Deliverables Submission and Approval

Ceer will have an online APQP deliverables submission and PSW e-Form available for suppliers within the Ceer Supplier Portal (Supplier Portal is under development). Suppliers will use Supplier Portal as the primary means of communicating and managing APQP deliverables with SQ.

Supplier Portal URL: TBD

Supplier Quality Activities Prior to Production Part Approval are described with specific roles and responsibilities as stated in CEER-QUSQ-SP5-L2-002 APQP Process Guide.



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Roles and Responsibilities

Suppliers can interface with Ceer representatives from the different Functions.

PUR = Procurement: Responsible for supplier identification and nomination, secures commitment for resource availability.

ENG = Engineering: Release engineering data; identify special characteristics and approve validation reports during APQP gate review.

SQS = Supplier Quality: Has overall responsibility for industrialization and driving APQP deployment at Supplier facilities during the new model launches.

SCM = Supply Chain Management: Responsible for scheduling and coordinating supply chain planning activities.

MLP = Manufacturing Engineering Logistics Packaging: Define packaging requirements.

SPR = Supplier: Responsible for applying APQP steps during VDP and producing objective evidence of deliverables.

#	APQP Deliverables	ENG	SQS	PUR	SCM	MLP	SPR
1	APQP Initiation	S	R	S	S	S	S
2	Procurement Contracts	S	I	R	I	I	S
3	Program Management	S	S	S	S	S	R
4	Feasibility Conformity	S	S	S	S	S	R
5	Engineering Records	A/R	S	I	-	-	R/S
6	Design FMEA	A/R	S	-	-	-	R/S
7	Special Characteristics	A/R	S	S	-	-	R/S
8	Validation Plan & Signoff	A/R	S	S	-	-	R/S
9	Traceability Conformance	S	S	S	I	I	R
10	Industrialization	S	A	S	I	I	R
11	Process Flow	S	A	-	-	-	R
12	Process FMEA	S	A	-	-	-	R
13	Process Control Plan	S	A	-	-	-	R
14	Measurement System	S	A	-	-	-	R
15	Dimensional Conformity	S	A	-	-	-	R
16	Appearance Approval	A	S	S	-	-	R
17	Supplier Verification Audits	S	A	I	I	I	R
18	Tiered Supply Governance	S	A	S	I	I	R
19	Packaging Agreements	S	S	S	I	A/R	R/S
20	Customer-Specific requirements	S	A	S	I	I	R
21	Safe Launch	S	A	S	S	S	R
22	Production Part Approval Process	S	A	I	S	S	R
23	Capacity Conformation	S	A	S	S	I	R

R = Responsible; **A** = Approve; **S** = Support; **I** = Informed; **C** = Consulted

Note: Wherever either or both roles are assigned, depending on the activity the secondary role shall be executed in addition to primary role.



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10. Production Part Approval Process PPAP

The supplier shall use PPAP Level 3 as the default level for all submissions unless otherwise specified by the Ceer SQS, as per AIAG Production Part Approval Process PPAP 4th Edition.

Regardless of the PPAP Submission Level, all the elements of a PPAP package should be completed according to Ceer PPAP Checklist (CEER-SQ-QM-001-F017).

Documents shall be submitted and updated via the “Ceer Supplier Portal” at the required due date.

For PPAP Levels and requirements at each program milestone (prototype and serial production vehicle build events), please refer to Appendix B: PPAP Submission and Approval Matrix, for PPAP Submission requirements. The supplier shall submit to the “Ceer Supplier Portal” all PPAP elements for interim and full approval as defined by program timing requirements.

I. PPAP Submission and Approval

During Product and Process Validation (PPV), an An Initial Submission Warrant (ISW) is required. It is used as proof that the supplier meets Ceer quality standards for samples delivered for PPV build event.

A Part Submission Warrant (PSW) is used from from Manufacturing Validation Build – Non-Saleable (MVB-nS) MRD to Start of Regular Production (SORP) and serial production. If any of the requirements does not meet the specification or is not fully released, please refer to the Deviations section below.

All parts and assemblies that are not “Full approved” shall be accompanied with PPAP deviation form showing, in detail, the nonconformances that prevent approval and the action plan. Any deviations to Final PPAP submission must be resolved prior to SORP.

II. Deviations

The deviation process shall be implemented in accordance with Appendix B PPAP Submission and Approval Matrix.

For all component deviations or non-conformances, the supplier must submit a Deviation Request Form (CEER-SQ-QM-001-F018) and a mitigation action plan via the “Ceer Supplier Portal” for approval.

Only upon approval of the deviation request form and mitigation action plan, the supplier shall ship the non-conforming components to Ceer Assembly Plant. All deviations shall also be documented in the APQP Open Issues List as per IATF 16949 guideline.

Rework shall be supported by written operating and inspection instructions. The inspection instructions shall be consistent with an updated production process control plan. Ceer will require special identification and segregation of the reworked product.

Material shipped under an approved deviation shall be labeled with the Deviation Request Form number provided by Ceer.



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11. Run at Rate – Supplier Capacity Verification

Run at Rates are conducted twice with suppliers providing self-assessment for percentage of share capacity for Ceer and the percent utilization against annualized Ceer awarded capacity according to CEER-SQ-QM-001-F024 Run at Rate and Supplier Capacity Verification Form. Based on Ceer component risk assessment, SQS may select specific parts of interest to participate in the Supplier's Run at Rate:

- **R@R #1:** 300 pcs minimum or four production hours (per Ceer SQS agreement) for the evaluation of machine cycle time shall be completed prior to MVB-S MRD and results shall be submitted to the "Ceer Supplier Portal".
- **R@R #2:** Conducted on minimum of eight working hours or significant serial production run, as agreed by Ceer SQS and Supplier. The objective is to verify that the installed capacity meets required parts per hour, through-put and the percentage utilization of installed equipment capacity is less than 80%. The supplier shall complete the run at rate evaluation prior to SSF- Start of System Fill and shall submit the results to the "Ceer Supplier Portal" prior to SORP as specified by Ceer SQS.

12. Safe Launch Plan (SLP)

The purpose of the SLP is for protection of the Ceer plant and end customer from quality spills and issues during critical phases of a program (i.e., pilot builds and launch) and to ensure that potential quality spills, which could arise are identified, contained, and corrected in the Supplier's manufacturing plant before shipment.

SLP applies to parts that meet any of the following criteria:

- New, modified, carryover, and service parts for new programs.
- Undergoing a post-launch engineering change or a process change.
- Undergoing a Resource (tooling /equipment moves and/or process changes / sub-supplier changes).
- Supplied from a process re-started after a duration of 60 days or more, unless otherwise specified by Ceer SQS.
- Other situations as mandated by the SQS (e.g., undergoing corrective action verification after a quality spill, etc.).

NOTE: Ceer may request SLP be initiated at any time, as required. SLP does not take the place of 3CPR (3rd Party Containment and Problem Resolution) / CSL (Controlled Shipping Level) but may be added as an additional level of control over those processes, if desired.

The SLP consists of controls, inspections, etc. that are over and above the standard production controls in the Supplier's Control Plan. It shall include, at a minimum, all the following characteristic types: Regulatory (including Safety, Emissions, Homologation, and Report), Functional, Important, and Critical and/or Special characteristics. Ceer may, at its discretion, add other characteristics as well.

Implementation of SLP activities is required of the Tier 1 suppliers as well as the sub-tier suppliers (Tier 2 and lower). Tier 1 suppliers must ensure their sub-suppliers implement an equivalent Safe Launch plan that includes an increased frequency of inspection, part retention, and continuing conformance testing. At a minimum, SLP shall begin with the first shipment of parts for the PPV build phase and end no earlier than the date of Start of Production + 90 days.

NOTE: Nonconformances detected on the product/process during the SLP shall be used to update the PFMEA, production Control Plan, Master PFMEA, and other quality documents, as applicable.



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13. Lot Traceability

All suppliers to Ceer shall have an effective lot definition and traceability procedure. The shipping number shall be linked to the lot traceability procedure in such a way that the delivered product, unless otherwise approved in writing by Ceer SQS, can be traced back to the raw material.

The maximum size of a lot shall consist of one shift or eight hours of production, whichever is smaller. The lot definition shall reflect all significant processes influencing the component/material with the shipping lot number reflecting the last value-added operation.

Suppliers shall ensure that their lot traceability system maintains its integrity throughout the entire extended supply chain, including raw material and purchased components/products.

The lifecycle of many components begins and ends within the facility of the supplier. There are those components, however, that require processing by outside companies to finish the process stream. These may include heat treatment, forging, coining, stamping, machining, coating, and other various special processes.

If the original lot were batch processed through the different secondary processes, there would be no need to change the original lot number. However, if the batch is split at a secondary processor, then the lot number for each of the batches should be unique.

Once manufacturing/assembly begins, a lot number changes if:

- One shift of production or eight hours is reached.
- A new lot of raw material is being used.
- The components undergo another value-added process, and the original lot is divided during processing.
- The lot number changes on any one of the components being used.
- When required, the supplier may need to implement:
 - Serialized (maintains a one-to-one relationship between the finished goods' serial number and the components' serial number) lot traceability; or
 - Specific Lot (maintains a one-to-one relationship between the finished good serial number and the components' lot numbers) traceability for certain programs.

To clarify the difference between this and general traceability, consider a supplier who stamps a given component. After stamping, two fasteners are then pressed into the stamping. General traceability is where there is no lot traceability between the stamped component and the assembled parts. Specific traceability would be where the lot numbers of the assembled components are traceable through the lot number of the stamped component.



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14. Special Characteristics

At a minimum, suppliers shall implement process controls for Significant, and Critical Characteristics as designated on Ceer drawings. Additional characteristics deemed germane to be “predictors of process stability” should also be identified in the supplier’s Control Plan.

These relate to product safety, government regulation, product performance and the ability to assemble product and/or customer satisfaction features. These are identified by various symbols requiring specific levels of special controls and process capability.

The supplier must calculate and report the process capability as Ppk. For those characteristics/features showing a Ppk of less than 1.67, the supplier must create an action plan that defines the containment and process improvements. Supplier to share the process capability performance to Ceer SQ upon request.

Process capability can be conducted with both variable and attribute data. The minimum acceptable sample size for variable data is 125 pieces and for attribute, 300 pieces, unless otherwise agreed with the Ceer SQS.

Either containment, 100% sorting, or some form of mistake proofing, must continue until such time that the process Cpk demonstrates capability greater than or equal to 1.33 unless otherwise agreed with the Ceer SQS.

To ensure this, capability reports must include a histogram, control charts and a normality test. Minitab is the preferred statistical software of choice. Please refer to the latest edition AIAG Manual on Statistical Process Control (SPC) and the Ceer PPAP Submission Checklist.

15. Pass-through characteristics (PTC)

Pass-through characteristics are defined as parts that are shipped to Ceer without being altered to form, fit or function to the safety-critical feature. Those characteristics are not controlled, or functionally tested anywhere downstream in the supply chain, are ultimately supplied to Ceer, and would have a significant impact on customer satisfaction and/or warranty.

Even if a PTC is not a Special Characteristic, Ceer's pass-through characteristic requirements adhere to the basic standards outlined in the AIAG CQI-19 Sub-Tier Supplier Management Process Guideline. To be considered, characteristics must have a PFMEA Severity greater than 4.

Tier I suppliers assume all responsibility for the quality of “pass through characteristics” that are considered safety critical. The supplier and Ceer must reach an agreement on the proper method of control for the identified PTC.

Where pass-through characteristics are identified, supplier shall define a process to:

- Classify and manage sub-supplier pass-through characteristics, utilizing error-proofing methods.
- Implement robust controls within the process control plans / work instructions.
- Demonstrate that pass-through characteristics are analyzed on all new parts and planned for current parts.
- Identify all pass-through characteristics on the sub-supplier's PFMEA and control plan / work instructions and integrate into the supplier's PFMEA and control plan.
- T1 suppliers with PTC must complete the Pass-Through Assessment Matrix



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16. Sub-tier Supplier Management

Ceer suppliers are responsible for the development of their sub-tier suppliers. Ceer suppliers shall have a defined process with appropriate resources to manage Tier 2/3 suppliers including the measurement of sub-supplier performance. Evidence of Tier 2 supplier PPAP completion and acceptance is required for all components and at a minimum, must include the PSW. Tier 2 supplier PPAP shall be completed prior to Tier 1 PPAP signoff.

Any change on Tier 2/3 suppliers shall be communicated in advance to Ceer via the “**Supplier Request for Change**” process for approval.

Ceer reserves the right to enter, participate in audits of Tier2/3 suppliers regarding quality management system, process, product, jointly with Ceer suppliers. Tier 1 suppliers should share the CQI audit results of Tier n's to Ceer SQ upon request.

17. Serial Production Introduction

Once Supplier Product and Process Development is complete, the Ceer SQS approves PPAP submission and Supplier Serial Production begins.

During this phase, there are several key requirements that suppliers shall assure compliance to. Key requirements include:

- Supplier reporting to Ceer for Issue Containment,
- Root Cause Identification,
- Corrective Actions,
- Supplier Request for Change,
- Sub-tier Supplier Management and Annual Revalidation.

18. Supplier Quality Problem Resolution (SQPR)

Suppliers are expected to immediately notify Ceer if suspect or known non-conforming material and/or finished goods have been shipped to Ceer as stated in the Ceer Supplier Quality Problem Resolution Procedure (SQPR).

Ceer shall use the CEER-QUSQ-SP5-L3-003 Supplier Quality Problem Resolution Procedure (SQPR) to immediately notify Suppliers of issues with Delivered Quality and / or systemic field product issues, where Ceer has identified nonconformity either through design or manufacture. Ceer recommendation is to follow a 8D methodology for problem solving.

After Ceer issues notification of a SQPR and provides a sample of defective parts as necessary, the supplier is required:

- To submit interim containment action (D3) within 24 hours after notification and update periodically.
- Root Cause Analysis, D4 needs to be carried out using appropriate problem-solving methods and submitted to Ceer. The detail of root cause analysis shall cover Why Happened? and Why Escape? and shall be submitted to Ceer within 10 working days after receipt of malfunction part, unless otherwise specified by Ceer SQS.



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- The completed 8D report, with permanent corrective action implemented, will be submitted within a total of 20 working days or specified by Ceer SQS.
- An Executive 3-Legged, 5-Why, 1-Pager requested by SQS.

Breakpoint 1: The first shipment of a product sent to Ceer after **D3 Containment Action** shall be clearly identified on and in the box.

Breakpoint 2: The first shipment of a product sent to Ceer after **D6 Permanent Corrective Action** implemented shall be clearly identified on and in the box.

Issue severity definition

Disruptions (issues) are classified in accordance with AIAG-VDA FMEA Severity definition.

The purpose of weighting discrepancies is to have a uniform method of prioritizing events which require fast corrective action.

The below stated descriptions are intended as a guideline for proper weighting assignment:

Issue Description	AIAG VDA FMEA Severity	Example
Walk home failures, legal and safety	9-10	Safety/Government Regulation label wrong/missing/wrong location, Defect related safety function (Airbag/Brake/Accelerator/Horn/ABS/Battery/Axle/Suspension/etc.) Fluid leak (Brake, Coolant, Power steering, washer), Incorrect Fluid, No fluid
Defects that affect basic performance of components and require a dealer to repair	7-8	Water leak, Fluid leak, Underbody damaged/function/missing parts/wrong part, Glass broken/crack/loose, HMI defect, Obvious Customer Concern, Incorrect routing (Safety), Any exposed wires
Defects that would cause major customer dissatisfaction	4-6	Surface/Appearance/Gap/Flushness depends on standard, Glass chip/Scratch, Continual Noise, Incorrect routing
Perceptual customer dissatisfiers	2-3	Surface/Appearance/Gap/Flushness depends on standard, Intermittent Noise
Minor customer dissatisfiers	1	Surface/Appearance/Gap/Flushness depends on standard.

Submission of Warranty Claim

Where Ceer and the Supplier both agree the Supplier has full or partial responsibility for the technical and, or systemic root cause of a systemic field defect, Ceer will issue a Warranty Claim to the Supplier, in accordance with “Ceer National Automotive Company, General Terms and Conditions for Direct Procurement of Goods” Section 12. For Supplier Warranty Claim Process, refer to CEER-QUSQ-SP5-L3-003.



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19. Supplier Request for Change (SRC)

This change notice process is applicable when suppliers request change for a Temporary or Permanent Process Change. These are typical manufacturing related processes and have no impact on design, appearance, fit, form, or function.

The suppliers shall submit a Supplier Request for Change (SRC) form via the “Ceer Supplier Portal” for Supplier Quality and Engineering approval. The Suppliers shall not implement any process related changes until SRC form approval is obtained from Ceer.

The change includes the requirement of tooling/equipment moving, new tooling or new equipment and any other process related changes.

Ceer reserves the right to chargeback the Supplier for any internal or external cost associated with the Supplier Request for Change such as, but not limited to Product Validation, New Supplier Assessment, Run at Rate, Launch Readiness Audit, PPAP Approval, Ceer travel and support costs.

Consequences of non-communicated or unauthorized process changes at the supplier’s manufacturing facility or any sub-supplier facility could result in any or all following actions:

- Written notification from Ceer to the supplier’s Registrar or from Ceer to the supplier, requesting they notify their registrar of the non-conformance.
- Consideration for New Business Hold (NBH) with defined exit criteria.
- Issuance of an A-rated QPRR with CS2 activated for immediate third-party containment (controlled shipment) of affected component/product.
- Potential request for independent, third-party audit of affected supply chain including all affected sub-tier suppliers involved.



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20. Annual Revalidation

The intent of annual revalidation is to ensure conformance to print, dimensional and capability requirements and all applicable Ceer specific requirements. Unless otherwise specified, a complete annual layout inspection including all sub-components is required for all components.

All suppliers shall have a plan to annually revalidate their respective production components. This includes all tools and cavities. Suppliers shall make this annual revalidation plan available to Ceer upon request.

At a minimum, on an annual basis, Ceer SQ shall identify suppliers required to submit evidence of revalidation for selected components. Notification, schedule, and submittal of revalidations identified will be submitted through the “Ceer Supplier Portal”.

Revalidation evidence submission requirements are product line specific, and criteria are based on component performance and categorization. Suppliers shall document this requirement in the Product Control Plan for all components supplied.

Unless a written agreement is complete at the original PPAP, the supplier should have a process to confirm ALL characteristics.

Features / characteristics / notes measured routinely as defined in the control plan are not required to be measured as a specific activity. However, in-process measurement results may need to be submitted, if required by Ceer.

Features / characteristics / notes not measured routinely as defined in the control plan must be measured and the results submitted.

If the annual revalidation reveals nonconformance to Ceer drawings, the supplier must immediately contact the Ceer responsible SQS and must supply the dimensional data and corrective action plans.

21. Supplier Facility Access

By prior notice, suppliers shall allow Ceer access to both their facilities and their supplier’s facility for the purpose of evaluating parts, processes, documents (i.e., FMEA, Control Plan, Instructions, records....), methodologies and systems used in the manufacturing of Ceer products.

Ceer may, at its discretion, use third-party independent auditors. These individuals represent Ceer and will audit the supplier's processes to establish conformance to validated quality systems.

22. Contingency Plan

It is the responsibility of the supplier to evaluate and identify all internal and external risks, and ensure robust contingency plans are created and maintained, as per, but not limited to, IATF 16949 guideline. These plans shall be made available to Ceer upon request.



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23. Document and Product Sample Retention

Suppliers shall retain documents and product samples for the time the part is active (a part is active if it is being supplied to the vehicle for original or service applications) in production plus a minimum period of 15 years at the end of serial production. Parts used on multiple programs may require an exceptionally long retention period.

The supplier shall retain a master sample from each cavity, die and pattern for the length of time that the component/material is active plus one year. The master sample / boundary shall be identified as such and shall show PPAP submission reference and Ceer approval date.

24. Ceer Properties – Ceer Tooling Asset

All tools, test, or inspection equipment belonging to Ceer will be permanently marked to clearly indicate Property of Ceer. These tools are to only be used for Ceer products unless an authorization in writing exists. Failure to comply with tool identification requirements will result in delay or non-payment.

Ceer will provide suppliers with the Tooling and Equipment Asset Tags to be affixed to all Ceer owned Tooling and Equipment.

25. Disposition of Nonconforming Products

Any component produced for supply to Ceer not sent directly to Ceer or an authorized third-party must be destroyed in-house prior to recycling such that the component may never be used in the intended application.

This includes scrap parts produced during production trials, engineering sampling, and all setup and inspection pieces. Only the PPAP samples are excluded from this policy.

Suppliers are expected to manage internal and sub-supplier inventories effectively to meet Ceer demand. Overproduction with intent to market components or distribute to unapproved entities is expressly forbidden.

Suppliers in violation of this policy, wittingly or unwittingly, will face significant penalties up to and including concern entry, new business hold, de-sourcing, or legal prosecution. All costs associated with non-compliance (including legal representation, court costs, and liabilities and damages) will be the sole responsibility of the supplier.

Suppliers will guarantee conformance to this practice and will guarantee all sub-suppliers will conform to this practice.

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26. Controlled Shipping Levels 1 and 2

Controlled Shipping (CS) Level 1 and 2 will be levied against the supplier when Ceer has determined that the supplier does not have the necessary safeguards preventing non-conforming products from reaching the Ceer Assembly Plant.

Controlled Shipping, Level 1 (CS1) is initiated by Ceer and performed at the supplier location by the supplier's employees. Controlled Shipping Inspection process must be performed in a controlled area of the plant. Secondary Inspection data must be collected, and the inspected product must be certified, and data is to be provided to Ceer.

Controlled Shipping, Level 2 (CS2) includes all of Level 1, with an added inspection by a Ceer approved 3rd party. The 3rd party is selected by the Supplier, approved by Ceer, and paid by the Supplier.

In some instances, Ceer may require the 3rd party inspection to be performed outside the supplier's facility. In all cases, the Supplier Quality Certification body must be informed by the supplier that they have been placed upon Controlled Shipping Level 2 by their customer and confirmation that this action has been completed must be provided to Ceer within 5 days.

Based on the severity of the incident, Ceer may elect to go directly to CS2, with Ceer Supplier Quality Director approval. The Ceer SQS will review irreversible corrective action and authorize removal or renewal of Controlled Shipping when appropriate.

NOTE: Minimum Record of 30 days Corrective Action verification period with no re-occurrences is mandatory.

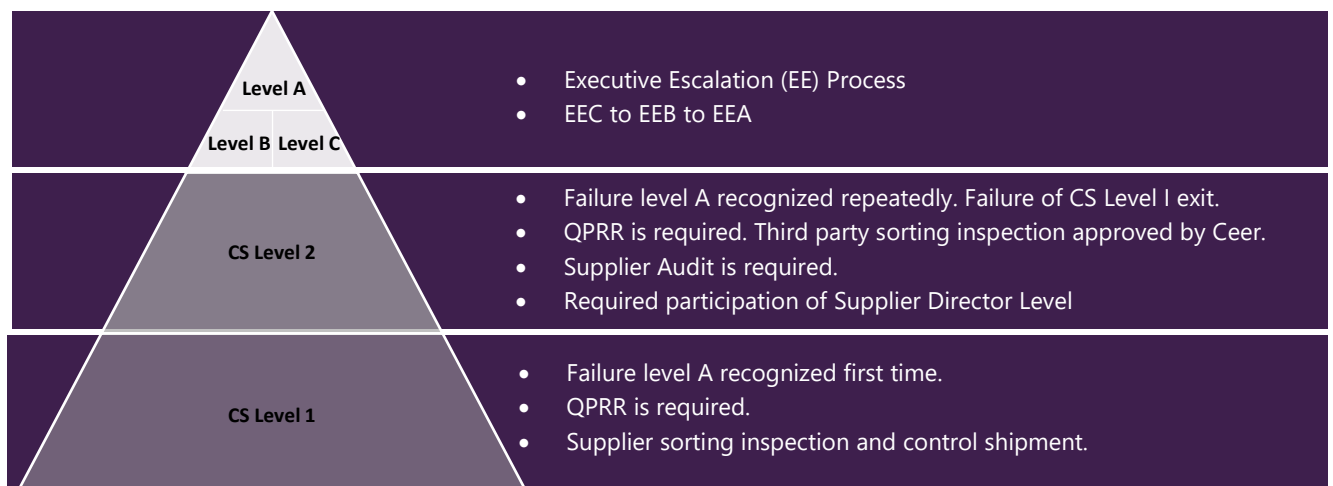


Fig.2. Control Shipping Process



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27. Executive Escalation (EE) Process

Suppliers providing Ceer with products and services that do not meet quality, delivery or planning commitments and requirements are subject to enrollment in the escalation process.

Executive Escalation (EE) is initiated by Ceer. The purpose of EE is to ensure continuous improvement of suppliers occurs with the best support of appropriate mentors.

EE classifies by three levels. In all levels, the Supplier takes responsibility to adapt Ceer requirement of data, evidence and information sharing to exit EE.

EE Level C (EE-C): Triggered by the supplier chronic and repeated poor performance such as non-conforming parts and delivery issues. Ceer will utilize a scorecard system to monitor supplier performance (*Under Development*).

EE Level B (EE-B): Triggered by the failure of EE-C exiting, and/or quality/delivery performance with continuing degradation.

EE Level A (EE-A): Triggered by the failure of EE-B exiting, and/or impact on vehicle launch and/or continuity of delivery quality.

In all stages of EE, the supplier can exit with adapting of the following exit criteria:

- Agreement reached to resolve issue.
- Three to six months improvement glide path.
- Monitoring and follow up plan for recurrence.

Ceer responsible SQS plays a role of leading suppliers in this activity and responsible for:

- Escalation Facilitator for successful event.
- Prepare the Supplier for success with pre-meeting guidance and slide deck review.
- Monitoring and meeting on Exit Criteria providing Executive Reports on progress.
- Prepare the Supplier De-escalation to Exit.



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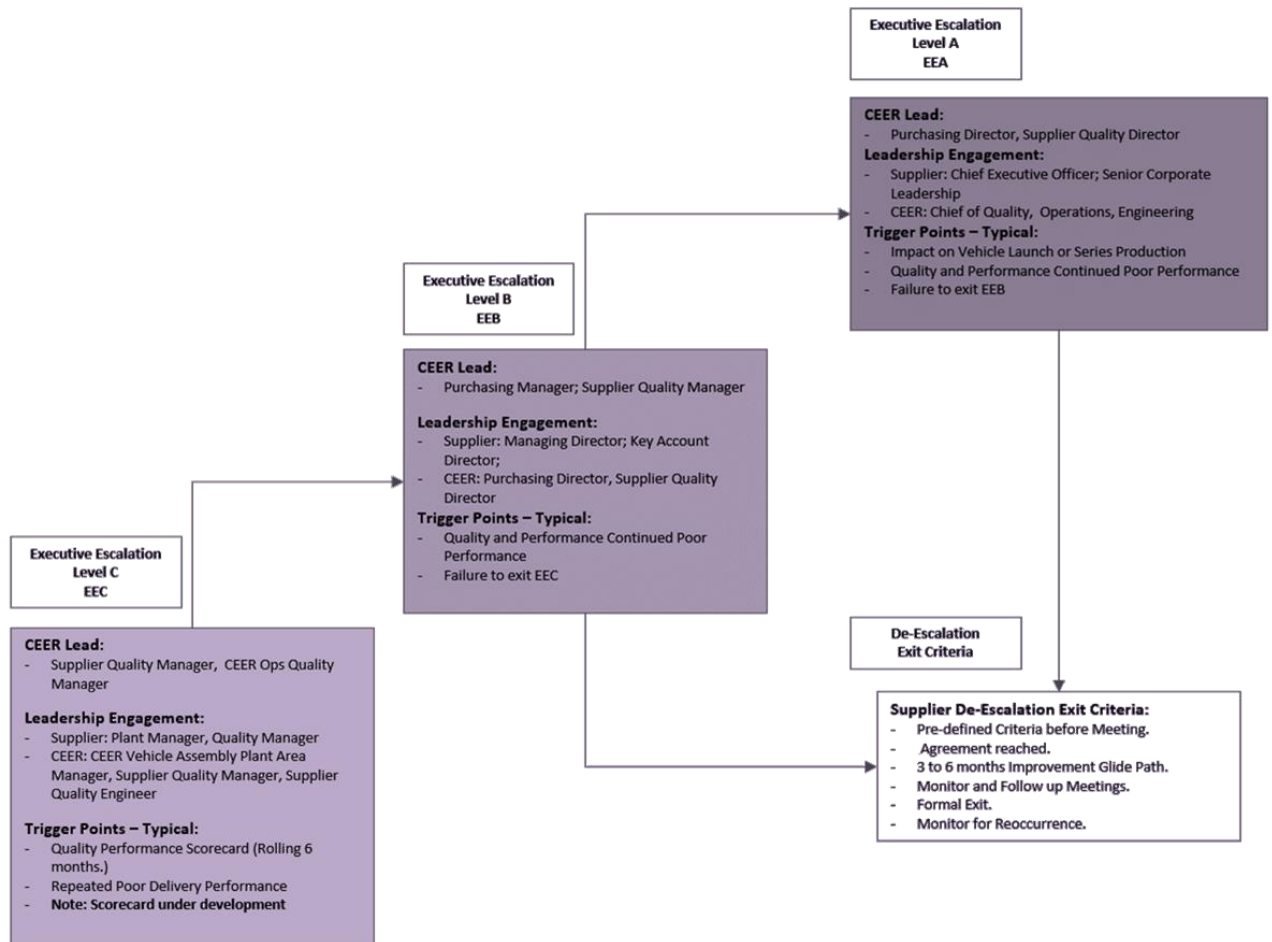


Fig.03. Executive Escalation Process



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28. Reference Document

- Automotive Industry Action Group (AIAG) Production Part Approval Process Reference Manual
- Automotive Industry Action Group (AIAG) Advanced Product Quality Planning Manual
- Automotive Industry Action Group (AIAG) Control Plan Manual
- Automotive Industry Action Group (AIAG) Failure Mode and Effects Analysis Manual
- Automotive Industry Action Group (AIAG) Statistical Process Control Manual
- Automotive Industry Action Group (AIAG) Measurement Systems analysis Manual
- QUSQ-SP5-L2-003 Potential Supplier Assessment (PSA)
- CEER-QUSQ-SP5-L3-003 Supplier Quality Problem Resolution Procedure (SQPR)
- CEER-QUSQ-SP5-L2-002 APQP Process Guide
- CEER-TEPI-COP2-L2-001 Ceer System & Design FMEA



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29. Appendix A: Acronyms and Abbreviations

Term	Definition
AIAG	Automotive Industry Action Group
APQP	Advanced Product Quality Planning
CC	Critical Characteristic
CP	Control Plan
CS	Controlled Shipping
DFMEA	Design Failure Mode and Effect Analysis
DV	Design Verification
EE	Executive Escalation
FOT	First Off-Tool Part
MSA	Measurement System Analysis
PFD	Process Flow Diagram
PFMEA	Process Failure Mode and Effect Analysis
PPAP	Production Part Approval Process
PPV	Product and Process Validation
PTC	Pass-through characteristics
SQPR	Supplier Quality Problem Resolution
RAM	Risk Analysis Matrix
SC	Special Characteristic
SLP	Safe Launch Plan
SPC	Statistical Process Control
SQ	Supplier Quality
SQPR	Supplier Quality Problem Resolution
SQS	Supplier Quality Specialist
TKO	Tooling Kick-Off

30. Appendix B: PPAP Submission and Approval Matrix

	Maturation Level	Build Phase	Pass-rate	Frequency	Other Requirements
PPAP Requirements	M1	Matching M1	80% of Critical points in spec to GD&T. - All other points within 150% of tolerance. (+/-1.0 tol. must be within +/- 1.5)	Minimum 5 parts measured for M1.	3D Scanning data required. - These builds are for Matching and ME try-outs. - Dimensional data required with M1 part delivery. - Parts off serial tools and fixtures / produced per launch control plan. - Reference Dimensional SOR for complete Dimensional requirements. Additional Requirements for parts to Body Shop: - Weld Strength (Nugget size, Position): 85%, Weld strength nuts: must meet torque spec, Stud weld strength: must meet torque spec. - Spotweld count: 100%, Stud weld count: 100%, Weld nut count: 100% - Weld Appearance (Distortion, Burrs, Spatter): 70% - Sealer and Adhesive: 80% to spec.
	PPV	PPV / Matching M2 (Product and Process Validation)	100% of Critical points in spec to GD&T. - All other points within 125% of tolerance/. (+/-1.0 tol. must be within +/- 1.25)	- Minimum 10 parts measured for PPV. - Minimum 3 parts measured for M2.	3D Scanning data required. - Dimensional data required with M2 part delivery. Initial PPAP Submission - Dimensional/Material/Function - Function verification complete and approved. - Parts off serial tools and fixtures / produced per launch control plan. Can be produced at alternate location. - All assemblies produced with serial production intent process. - Initial IMDS submission. Additional Requirements for parts to Body Shop: - Weld Strength (Nugget size, Position): 90%, Weld strength nuts: must meet torque spec, Stud weld strength: must meet torque spec. - Spotweld count: 100%, Stud weld count: 100%, Weld nut count: 100% - Weld Appearance (Distortion, Burrs, Spatter): 80% - Sealer and Adhesive: 100% to spec

	Maturation Level	Build Phase	Pass-rate	Frequency	Other Requirements
PPAP Requirements	Non-Saleable	MVB-nS (Manufacturing Validation Build – Non-Saleable)	100% of All points in spec to GD&T	Minimum 10 parts measured.	3D Scanning data required. Non-Saleable PPAP Submission - Parts off serial tools and fixtures / produced per launch control plan. - All parts and assemblies produced at serial location. - Material must be within spec. 100% Grained Parts Required. - Full component/sub-system PV (function and durability) completed. - Parts meet AAR requirements. Additional Requirements for parts to Body Shop: - Weld Strength (Nugget size, Position): 95%, Weld strength nuts: must meet torque spec, Stud weld strength: must meet torque spec - Spotweld count: 100%, Stud weld count: 100%, Weld nut count: 100% - Weld Appearance (Distortion, Burrs, Spatter): 90% - Sealer and Adhesive: 100% to spec
	Saleable	MVB-S (Manufacturing Validation Build – Saleable)	100% of All points in spec to GD&T	Refer to PPAP standard.	3D Scanning data required. Saleable PPAP Submission - Parts off serial tools and fixtures / produced per launch control plan. - Any deviation to dimensional specs must be approved by CEER prior to shipping according to PPAP procedure. - No rework parts allowed. - Final AAR / Final Grained Parts. - Component and Sub system Process Validation complete. - All parts and assemblies produced at serial location. - Final IMDS submission complete and approved. Additional Requirements for parts to Body Shop: - Weld Strength (Nugget size, Position): 98%, Weld strength nuts: must meet torque spec, Stud weld strength: must meet torque spec. - Spotweld count: 100%, Stud weld count: 100%, Weld nut count: 100% - Weld Appearance (Distortion, Burrs, Spatter): 98% - Sealer and Adhesive: 100% to spec.
	Full approval	SORP (Start of Regular Production)	Refer to PPAP standard	Refer to PPAP standard.	- Full PPAP achieved per program timing (Ref PPAP standard). - Any deviations to Final PPAP submission must be resolved prior to SORP

31. Appendix C: A-B-C-D Sample Definition

	Category	Build Phase	Definition	Maturity Level	Requirements
A-B-C-D Samples Definition	A – Sample	Mule Build	Functional samples for prototypes, usually with limited drivability and low degree of maturity.	Mule Samples	- Basic SW functions under test conditions. - Refer to the Ceer Statement of Work (SOW) and Engineering's part specifications.
	B – Sample	PT (Prototype Build)	Functional samples with Full drivability and high degree of maturity.	Prototype Samples (PT)	- All functions are available and installation dimensions are known. - All SW functionality as required. - Refer to the Ceer Statement of Work (SOW) and Engineering's part specifications.
	C -Sample	PPV (Product and Process Validation Build)	Fully functional samples manufactured with series production tools, under pre-series condition manufacturing processes.	Off Tool Samples (OTS)	Refer to Supplier Quality Handbook - Appendix B: PPAP Submission and Approval Matrix.
	D- Sample	MVB - S (Manufacturing Validation Build – Saleable)	Fully functional samples manufactured with series production tools, under series condition manufacturing processes.	PPAP Samples	Refer to Supplier Quality Handbook - Appendix B: PPAP Submission and Approval Matrix.
	Series Production	MVB-S + (Manufacturing Validation Build – Saleable) +	Serial parts with PPAP status full approved.	Full Approval	Refer to PPAP standard.



CEER SUPPLIER QUALITY HANDBOOK

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32. Appendix D: Accessories

Definition

Additional components or features installed in a vehicle to enhance its functionality, comfort, safety, performance, or aesthetics.

Marketing enabler to provide customized variants /upgrade options to improve the overall customer experience and pride of ownership.

Categorization

- Off-The-Shelf (OTS): Accessories are already available in the market and retailed as supplier branded.
- Ceer Designed (BTP): Accessories are fully designed by CEER and manufactured as BTP by suppliers.
- Supplier Designed (FSS): Accessories are designed/modified by supplier with specific requirements from CEER.

Requirements

Collaboration throughout development and manufacture is critical for ensuring the quality of automotive accessories for Ceer Motors Company.

Accessory parts suppliers and service providers must be ISO9001 or equivalent certified, with IATF16949 certification being mandatory depending on the nature of the accessories, as agreed with Ceer SQS.

Testing and certification, whether in-house or third-party, are required to validate performance and safety. Fit and compatibility are necessary, enabling seamless integration without alterations. Robust supplier quality management systems must be in place, encompassing selection, inspection, and control procedures. Traceability of materials and components plays an essential role in quality assurance and compliance. Clear labeling on packaging is required to secure it during travel and installation. Continuous improvement activities drive enhancements to product design and production processes. This comprehensive strategy guarantees that accessories meet Ceer requirements and fosters long-term cooperation.